Metabolite Interaction Analysis

Donguk Shin

Seoul National University
Department of Statistics

Main Objectives and Outline

Main Objectives

- To investigate whether interactions among plant metabolites influence insect performance.
- To gain preliminary insights that may guide future research.

- Introduction
- Analysis Results
- Discussion

Data

Data Structure:

- The dataset consists of:
  - ▹ Insect performance (response, Y).
  - ▹ Metabolite abundances (covariates, X).
  - ▹ Four groups (`le, mp, px, sl`).
- Each group contains 24 observations, resulting in a total of 96 samples $(n = 96)$.
- X comprises 4,904 variables $(p = 4904)$, including a control group.
- Since this is an ultra-high-dimensional dataset $(n \ll p)$, variable screening is essential.

Data

Data Preprocessing:

- Because the response is positive, a log-transformation was applied:

$$y_i \leftarrow \log(y_i).$$

- Using the control group, metabolite abundances were transformed into log2FC:

$$x_{ij} \leftarrow \log_2 x_{ij} - \log_2 x'_i,$$

  where x'_i denotes the corresponding value from the control group.

- Subsequently, group-wise normalization was performed.

Data

	pf	group	X1285	X4560	X7320	X7322	X4553	X853	X857	X3919	X1330	X64
1	0.746081870	le1	-0.95695728	-0.147428429	-1.507062868	-0.2505312721	-0.35152229	-0.469457736	-0.451009902	-0.08182775	-0.01723794	-0
2	-1.211716049	le1	1.06577949	-0.172890693	0.347087638	0.6643493546	-0.40751500	0.538054180	0.691126490	1.04981264	-1.32936363	-1
3	0.385377708	le1	-0.64630437	-0.157296634	0.793248747	0.5696389786	-0.37322294	0.654701949	0.428402305	0.32703446	-0.52576796	0
4	1.090756218	le1	-1.02565439	-1.833260307	0.010157742	0.4818749488	-0.30000983	1.059476655	0.963414142	0.15676436	1.18989824	0
5	0.867190576	le1	0.26073186	-0.122138506	1.795070649	-2.3727834115	-0.29590857	-0.187367958	-0.133533289	-2.16725356	1.28600664	1
6	-0.835251191	le1	1.63523432	1.052275034	-0.388337361	0.2587492530	-0.33894179	0.687335397	0.675204532	0.20069765	0.27757187	0
7	0.385377708	le1	0.41724195	-0.172001003	-0.690063487	0.4502924331	-0.40555853	-0.185863949	0.002057086	0.92443494	-1.28351602	-0
8	-1.427816841	le1	-0.75007158	1.552740538	-0.360101060	0.1984097155	2.47267895	-2.096878538	-2.175661363	-0.40966274	0.40240880	-0
9	-0.343262820	le2	0.56555866	-0.286623412	0.673827455	0.2033462381	0.28962738	-1.033772996	-0.358508475	0.09596133	-0.34069430	-2
10	1.553254469	le2	0.64200100	1.047538164	0.752929829	0.6491484985	-0.49590769	0.969926157	0.589892433	0.40477691	-0.36539379	0
11	-1.251840731	le2	-0.74506266	1.100185645	0.280395025	0.5343762732	-0.98087781	-0.884593297	-1.419682572	-0.02700231	-0.38064266	0
12	-0.343262820	le2	-0.52637951	-1.880868007	0.189654024	0.4033266770	1.99016335	1.577362420	1.500278823	-2.38313283	-0.28722452	0
13	0.894050429	le2	-1.34462431	0.901574558	-1.007330017	-0.0116570258	-0.30870962	-0.406979837	-1.258822797	0.06460210	-0.35950774	0
14	-1.251840731	le2	1.38524573	-0.296971237	-0.651572661	-2.3452690229	-0.35340084	0.730342408	0.460282999	0.66342889	-0.36091297	0

Methods – Screening

Screening: A HSIC-based screening method was applied separately within each group.

1. 25 variables were selected using HSIC-ordering (Song et al., 2007).

$$\mathrm{HSIC}(Y, X_j) = 0 \Longleftrightarrow Y \text{ and } X_j \text{ are independent.}$$

2. The HSIC-Lasso (Yamada et al., 2014) was applied to extract the top 5 variables.

$$Z = (Z_1, Z_2, \ldots, Z_5)^\top, \quad \text{denoting the top five indices with } \hat{\alpha}_j > 0,$$

$$\widehat{\boldsymbol{\alpha}} = \arg\min_{\boldsymbol{\alpha} \geq \mathbf{0}} \left\{ \frac{1}{2} \sum_{j=1}^{25} \sum_{k=1}^{25} \alpha_j \alpha_k \,\mathrm{HSIC}(X_j, X_k) - \sum_{j=1}^{25} \alpha_j \,\mathrm{HSIC}(Y, X_j) + \lambda \sum_{j=1}^{25} |\alpha_j| \right\}. \tag{1}$$

Methods – Detection

Interaction Detection: Interactions were heuristically identified using three approaches.

- **Random Forest–based approach:** Minimal Depth Interaction Detection (MDID, Paluszynska et al., 2020).
- **Penalized linear regression approach:** hierNet (Bien et al., 2013).

$$\min_{\beta,\Theta} \frac{1}{2n}||y - X\beta - X\Theta X^\top||_2^2 + \lambda_1||\beta||_1 + \lambda_2||\Theta||_1,$$

  - ▹ Θ automatically enforces heredity conditions (interactions only included if main effects exist).

- **Penalized Quadratic linear regression approach:** sprinter (Yu et al., 2020).

$$\min_{\beta,\gamma} \frac{1}{2n}||y - X\beta - Z\gamma||_2^2 + \lambda_1||\beta||_1 + \lambda_2||\gamma||_1.$$

  - ▹ Γ includes quadratic main-effect terms.

- For both penalized methods, the selected linear models were refitted using OLS.

Results – Group `le`

Variable Selection:

$$(Z_1, Z_2, Z_3, Z_4, Z_5) = (X_{447}, X_{1208}, X_{648}, X_{2677}, X_{4325}).$$

Interaction Detection:

- **MDID:** The pairs $Z_3 : Z_4$ and $Z_4 : Z_5$ exhibited low minimal depth.
- **hierNet:** $Z_3 : Z_4$ was significant.
- **sprinter:** $Z_3 : Z_4$ was significant.

Conclusion: For Group `le`, the interaction $Z_3 : Z_4$ is likely to be important.

Results – Group `le`

Table 1: ANOVA table of hierNet - Group `le`

Coefficient	Estimate	Std. Error	t value	$Pr(> \|t\|)$
(Intercept)	-0.041308	0.113490	-0.364	0.72220
Z_1	0.496185	0.209463	2.369	0.03548^{*}
Z_2	-0.331858	0.151928	-2.184	0.04951^{*}
Z_3	-0.348984	0.097236	-3.589	0.00372^{**}
Z_4	0.224014	0.128282	1.746	0.10629
Z_5	0.007718	0.172123	0.045	0.96497
$Z_1 : Z_4$	0.287283	0.146939	1.955	$0.07426^{\cdot}$
$Z_1 : Z_5$	0.111601	0.124601	0.896	0.38805
$Z_2 : Z_3$	0.287710	0.175017	1.644	0.12612
$Z_2 : Z_5$	-0.109263	0.115374	-0.947	0.36230
$Z_3 : Z_4$	0.465451	0.124587	3.736	0.00284^{**}
$Z_3 : Z_5$	0.243813	0.156023	1.563	0.14410

Results – Group `mp`

Variable Selection:

$$(Z_1, Z_2, Z_3, Z_4, Z_5) = (X_{1043}, X_{4679}, X_{3815}, X_{4020}, X_{4049}).$$

Interaction Detection:

- **MDID:** $Z_2 : Z_5$ was significant.
- **hierNet:** No interactions detected.
- **sprinter:** No interactions detected.

Conclusion: For Group `mp`, no metabolite interactions were detected.

Results – Group $\mathtt{px}$

Variable Selection:

$$(Z_1, Z_2, Z_3, Z_4, Z_5) = (X_{807}, X_{3001}, X_{772}, X_{37}, X_{5855}).$$

Interaction Detection:

- **MDID:** $Z_1 : Z_5$, $Z_1 : Z_2$ were significant.
- **hierNet:** $Z_2 : Z_4$, $Z_2 : Z_5$ were significant.
- **sprinter:** $Z_2 : Z_4$, $Z_2 : Z_5$ were significant.

Conclusion:

- The MDID results differ from those of the linear models.
- In both linear model solution paths, $Z_2 : Z_4$ and $Z_2 : Z_5$ consistently appear.

Results – Group px

Table 2: ANOVA table of hierNet - Group px

Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
(Intercept)	-0.003966	0.192006	-0.021	0.98386
Z_1	0.098317	0.560550	0.175	0.86369
Z_2	-0.506488	0.184692	-2.742	0.01785*
Z_3	-0.296199	0.551890	-0.537	0.60128
Z_4	0.090038	0.168415	0.535	0.60268
Z_5	-0.120902	0.175937	-0.687	0.50503
$Z_1 : Z_2$	-0.284262	0.319582	-0.889	0.39123
$Z_1 : Z_3$	-0.029170	0.186987	-0.156	0.87863
$Z_2 : Z_4$	0.421320	0.171970	2.450	0.03060*
$Z_2 : Z_5$	-0.774955	0.213495	-3.630	0.00345**
$Z_3 : Z_5$	-0.121759	0.229234	-0.531	0.60500
$Z_4 : Z_5$	-0.362881	0.218179	-1.663	0.12214

Results – Group `sl`

Variable Selection:

$$(Z_1, Z_2, Z_3, Z_4, Z_5) = (X_{1048}, X_{995}, X_{7309}, X_{3755}, X_{8157}).$$

Interaction Detection:

- **MDID:** $Z_3 : Z_4$, $Z_3 : Z_5$ were significant.
- **hierNet:** $Z_4 : Z_5$ was significant.
- **sprinter:** No interactions detected.

Conclusion: For Group `sl`, no consistent metabolite interactions were detected.

Discussion

Conclusions:

- For Group `le`, the interaction $Z_3 : Z_4$ appears potentially significant.
- For Group `px`, the interactions $Z_2 : Z_4$ and $Z_2 : Z_5$ appear potentially significant.

Limitations:

- The current methods lack rigorous statistical guarantees and rely on heuristics.
- The screening and selection procedures are not independent, leading to potential selection bias.
  - ▹ A post-selective inference framework that accounts for interactions is needed.

Suggestions:

- The sample size is extremely limited; additional data collection is necessary.
  - ▹ Insights from this analysis can guide more efficient sample acquisition.
- There is a need to develop post-selective inference methods that explicitly incorporate interactions.

Reference

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